The "hence" is invoking a standard principle — that the Taylor coefficients of a function (the coefficients of its expansion about 0) are controlled by its singularity nearest the origin (the *dominant singularity*) [2, 5]. Let me unpack why that principle applies and what it gives, because it is really two claims stacked together, one elementary and rigorous, the other standard but leading-order.

Setup. The g_m are the Taylor coefficients of G about the origin: $G(t) = \sum_{m\geq 0} g_m t^m$. The preceding sentence established that G's singularities (the places where this series stops representing G) are the zeros of H, the nearest being t_0 at distance $\rho = |t_0| \approx 0.043$.

Which Taylor expansion "the" refers to. By "the Taylor coefficients" I mean the coefficients of the expansion about 0 — the g_m in $G(t) = \sum_{m\geq 0} g_m t^m$. A function has a different Taylor expansion about every point at which it is analytic, so the phrase is meaningful only once a center is fixed; here the center is 0, and I should have said so. Three reasons make 0 the center that matters, in increasing order of importance.

(i) It is the expansion we are after. G is defined as a power series in t about the origin: the $g_m = h_m^{(0)}$ are produced by the recurrence (D) from the seed $g_0 = 1$, which is exactly the expansion at 0. The deformed coefficients $h_n^{(c)} = \sum_k \frac{c^k}{k!} g_{n-k}$, the characteristic polynomials, and the minimum moduli are all built from these g_m. No other expansion enters the problem, so "the Taylor coefficients" can only mean these.

(ii) It is forced by the normalization. The sequences are pinned at $h_0 = 1$, i.e. $G(0) = 1$; the generating function is by construction organized about $t = 0$. Expanding about another point would not produce the sequence under study.

(iii) The radius — and hence which singularity is "nearest" — is defined relative to the center. The expansion about 0 converges in the largest disc *centered at* 0 containing no singularity; its radius is the distance from 0 to the closest zero of H, namely $\rho = |t_0|$ [1, 2]. This is what makes t_0 the nearest singularity, and what ties $|g_m|^{1/m} \to 1/\rho$ to t_0 specifically. The phrase "nearest the origin" in the main document already encodes the choice of center: nearest *to* 0. Expanding about some other point a would recenter the disc at a, make "nearest" mean nearest to a, change the radius to the distance from a to the closest zero, and govern those (different) coefficients by a possibly different zero.

So the unsloppy phrasing is *the coefficients of the Taylor expansion of G about* 0, and it is the one that matters because it is the very sequence g_m the problem is made of — and the growth law $|g_m|^{1/m} \to 1/\rho$ is meaningful only once the center (here 0) fixes what "radius" and "nearest singularity" refer to.

Layer 1 — the growth rate (rigorous, elementary). A power series converges exactly up to the nearest singularity of the function it represents;

its radius of convergence equals the distance to that singularity [1, 2]. So the radius of $\sum g_m t^m$ is ρ. By the Cauchy–Hadamard formula (radius = $1/\limsup_m |g_m|^{1/m}$) [1], this is the same as saying

$$|g_m|^{1/m} \longrightarrow \frac{1}{\rho}, \qquad \text{i.e.} \qquad |g_m| \text{ grows like } \rho^{-m}.$$

That much is airtight, and it is exactly what the growth-rate figure of the companion note confirms ($|g_m|^{1/m} \to 23 \approx 1/\rho$). So the *modulus* of the nearest zero of H sets the exponential growth rate. Already this is one sense of "governed by the nearest zero."

Layer 2 — why the nearest one, and the phase (standard, leading-order). Why does the *nearest* singularity dominate, rather than all of them mattering equally? Because any singularity at a point t_k contributes a term of size $\sim t_k^{-m}$ to the m-th coefficient [4, 5]. (A simple example: a pole, $\frac{1}{t_k - t} = \frac{1}{t_k}\sum_m (t/t_k)^m$, has coefficients t_k^{-m-1}; a branch point $(1-t/t_k)^\alpha$ has coefficients $\sim t_k^{-m} m^{-\alpha-1}$. In both cases the geometric factor is t_k^{-m}.) So heuristically

$$g_m \;\approx\; \sum_k c_k\, t_k^{-m} \times (\text{slowly varying}),$$

and the term with the *smallest* $|t_k|$ wins, because $|t_k^{-m}| = \rho_k^{-m}$ is largest when ρ_k is smallest. The next zero sits at $\rho_2 \approx 0.078$, so its contribution is smaller than the nearest by a factor $(\rho/\rho_2)^m = (0.043/0.078)^m \approx 0.56^m$ — already negligible by $m \approx 20$. That exponential suppression of everything but the nearest singularity is the content of "hence."

Now the *phase*. Near a simple zero t_0 of H, a short calculation gives $J = tH'/H \approx t_0/(t-t_0)$ [1], so $\int J \approx t_0 \log(t-t_0)$ and $G \approx (t-t_0)^{t_0}$ — a branch point. Its coefficient contribution carries the geometric factor t_0^{-m}, and since $t_0 = \rho e^{i\theta}$,

$$t_0^{-m} = \rho^{-m} e^{-im\theta}.$$

The modulus ρ^{-m} is the growth (Layer 1); the unit-modulus part $e^{-im\theta}$ is a rotation by angle θ each step. Because H has real coefficients its zeros come in conjugate pairs $t_0, \bar{t}_0$, and adding the two conjugate contributions turns $e^{\pm im\theta}$ into a real cosine:

$$g_m = C\,\rho^{-m}\cos(m\theta + \varphi) + O\big(\rho_2^{-m}\,\mathrm{poly}(m)\big), \qquad \rho_2 > \rho,$$

with ρ_2 the modulus of the next singularity, so the omitted terms are smaller by the geometric factor $(\rho/\rho_2)^m \to 0$ (a slowly varying algebraic factor on the main term is suppressed here). The relation gives the dominant asymptotics, with everything else of strictly smaller exponential order; it is not a statement that the ratio of the two sides tends to 1, which would fail at the zeros of the cosine. So the *argument* θ of the nearest zero sets the oscillation [3, 5]. That is the second, and for us the important, sense of "governed by the nearest zero."

So the "hence" says. Of all the branch points of G (= zeros of H), only the nearest matters for large m, because the others are exponentially smaller; and that nearest one delivers both pieces of the asymptotics — its modulus ρ the growth rate ρ^{-m}, its argument θ the oscillation $\cos(m\theta + \varphi)$. Everything downstream (the period-4 in μ_n, the ≈ 37 envelope) is read off θ.

Honest qualification. Layer 1 is rigorous. Layer 2 — that the nearest singularity's *local type* fixes the full leading form including the polynomial factor and phase — is the transfer principle of singularity analysis (Flajolet–Odlyzko) [4, 5], which is a theorem under regularity hypotheses: the singularity should be isolated and of tractable (algebraic/logarithmic) type, with G continuable in a neighborhood beyond the disc (a Δ-domain). Those hold plausibly here — t_0 is isolated, well inside the unit disc, and a genuine branch point — and the prediction is confirmed numerically, but I have not verified the technical hypotheses for this particular G (in part because the behavior of H out on $|t| = 1$, where the $\tau(p_n)$ live, is not something I control). So treat Layer 2 as "the standard principle, leading order, empirically confirmed," not as a proved theorem.

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